

Multilingual Voice Assistant

Mallidi Akhil Reddy
2024TM93056

Multilingual Voice Assistant

Overview of a
React-based
assistant that
uses:

- Speech-to-text

- Intent
detection

- Llama
(Ollama)
responses

How It Works



1. User speaks



2. Browser converts
speech → text



3. Text sent as
prompt



4. Llama model
generates reply



5. Reply shown as
text



6. Optionally
spoken back

Core Features



✓ SPEECH
RECOGNITION



✓ TEXT INPUT



✓ INTENT DETECTION
(TIME, TRANSLATION,
WEATHER)



✓ LLAMA 3.2:3B
LOCAL INFERENCE



✓ DYNAMIC VOICE
OUTPUT



✓ LANGUAGE
SWITCHING
(EN/HINDI)

Speech Recognition Flow

- Uses Web Speech API (SpeechRecognition)
- Converts spoken audio → text
- Auto-fills text box
- Sent to assistant like a normal message

Assistant Processing Logic

- Detects intents (time, weather, translate)
- Extracts entities (city, language)
- If model call needed → sends to Ollama
- Manages conversation context
- Sends reply to UI

Text-to-Speech

- Uses speechSynthesis API
- Hindi & English voices
- Speaks assistant's response
- Queue-based speech processing

Translation Flow

1. User: 'Translate hello to Hindi'

2. Detects translation intent

3. Sends phrase to Llama

4. Llama returns direct translation

5. Shown + optionally spoken

Weather Flow

- Extracts city name from user text
- Uses OpenWeatherMap API
- Displays temperature + conditions

Tech Stack

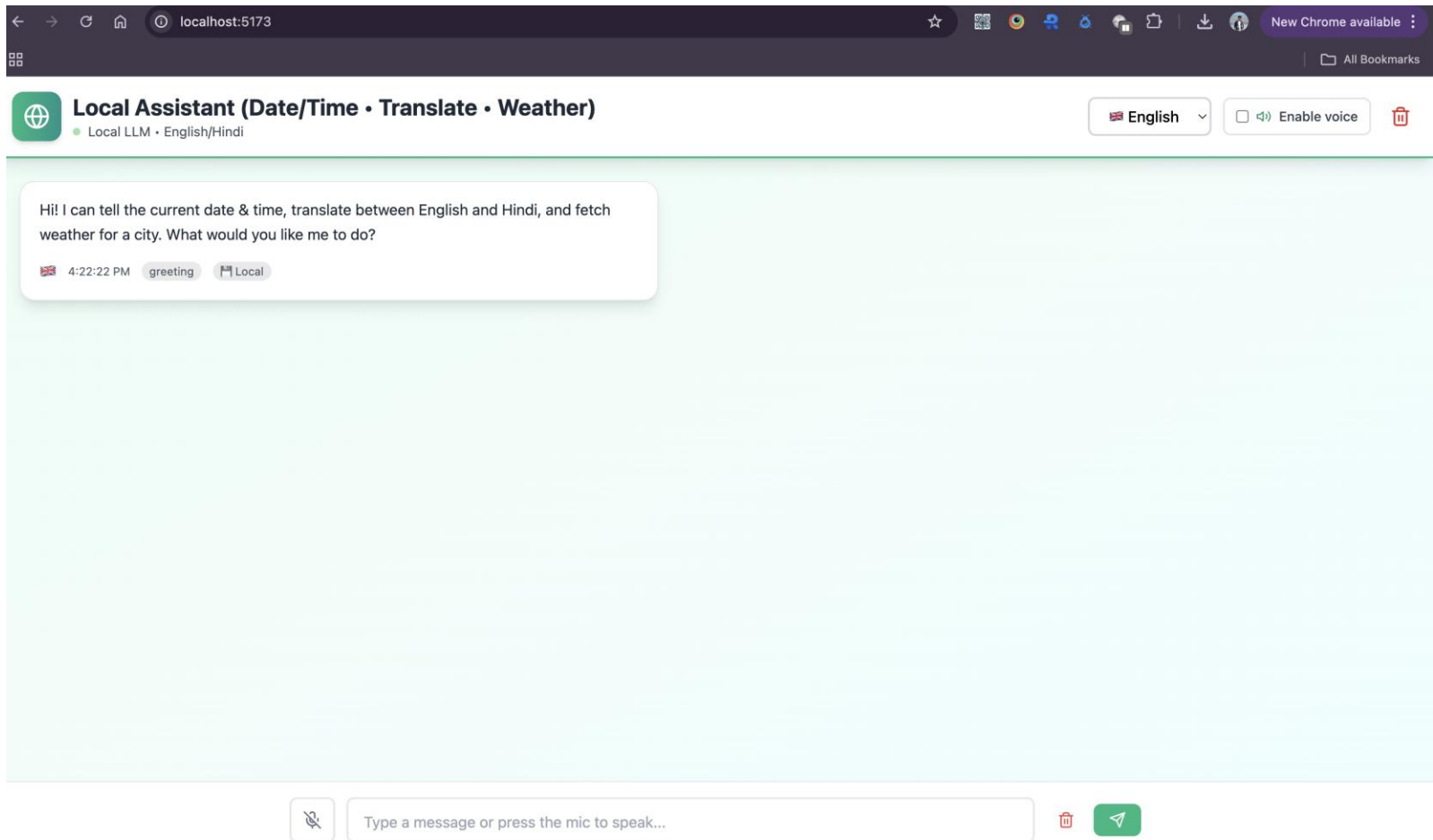
Frontend: React, TailwindCSS, Web Speech API

Backend: Local Ollama LLM

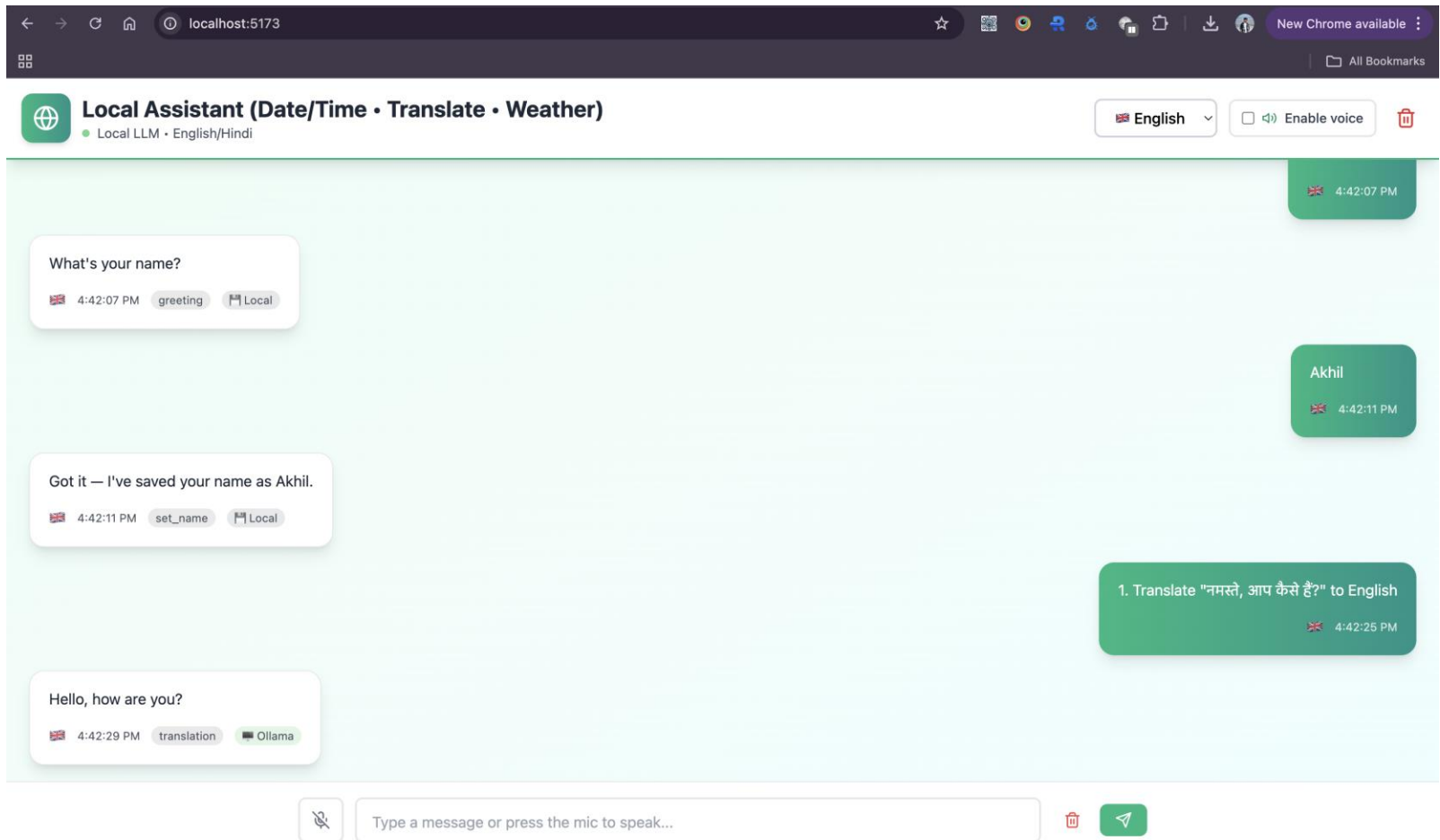
APIs: OpenWeather

Voices: speechSynthesis

Screenshots



Screenshots



Github URL

- <https://github.com/itzzmeakhi/voice-assistant>



Thank you